Natalie Block

Valdosta, GA | 832-525-9850 | natalie.rblock@gmail.com | LinkedIn | GitHub | Personal Website

Biostatistician and Statistical Analyst

Statistician proficient in R (tidyverse, Shiny, ggplot2), Python (pandas, scikit-learn, PyTorch), and SAS. Experienced in regression modeling, survival analysis, machine learning techniques, command-line tools, and building interactive dashboards and visualizations. What I enjoy most about my work is uncovering underlying patterns in data and translating them into meaningful insights.

Education

Master of Science in Public Health in Biostatistics, Emory University (May 2026)

Bachelor of Science in Cell and Molecular Biology, Texas Tech University (May 2024)

Core Competencies & Technical Skills

Core Competencies: Data Visualization, Dashboard Design, Statistical Analysis, Big Data, Machine Learning, Automation, Data Cleaning

Technical Skills: R, Python, Git, Bash, SAS, Excel

Technical Projects

CHIP and Inflammation Dashboard
Project Link | Completed: April 2026
Technologies: Shiny Dashboard, R, Excel

- Developed a comprehensive dashboard using R for analysis and Shiny for data visualization, displaying the estimated effect of genetic mutations on human inflammatory proteins.
- Reduced model analysis and visualization time from ~3 days to ~1 day by building fully reproducible workflow integrating Bash and R scripts

Machine Learning Comparison in Python
Project Link | Completed: In Progress
Technologies: Python

- Use scikit-learn and PyTorch to compare machine learning (ML) techniques' ability to classify outcomes. Includes convolutional neural networks, natural language processing, logistic regression, random forests, and XGBoost.

Professional Experience

Conneely Lab, Emory University ***September 2024-Present***
Graduate Research Assistant

- Analyze large-scale population cohort (n≈500,000) using R and Bash
- Clean data, fit mixed effects and machine learning models, and run model diagnostics
- Develop publication visualizations using ggplot2 and build interactive Shiny dashboards
- Review and implement statistical methods from literature to improve model
- Collaborate with geneticists, statisticians, and computational biologists to design and test self-driven hypotheses on genetic variants' effects on protein abundance

Involvement & Awards

REAL Program Award Recipient, Emory University
Dean's Council Merit Scholarship, Emory University
SISG Teaching Assistant, Epi & Gene Regulation
Poster Presentation, CHARGE North Carolina Meeting